

Sec. 2.1 Mathematical definitions

★ What is an “integer”?

Definition 1. Let n be an integer. Then

- n is **even** if
- n is **odd** if
- The **parity** of an integer is whether the integer is odd or even.

Examples:

Definition 2. A number x is **rational** if

Examples:

Definition 3. Let x, y be two integers. Then x **divides** y if and only if and for some integer k . (Equivalently, “ y is divisible by x ”, “ y is a multiple of x ”, “ x is a factor of y ”, “ x is a divisor of y ”.)

Notation:

Definition 4.

- An integer $n \geq 2$ is called **prime** if and only if the only (positive) divisors of n are 1 and n .
- An integer $n \geq 2$ is called **composite** if and only if it is not prime, that is,

Examples:

Properties of 0

- Parity:
- Rationality:
- Divisibility:
- Primality:

Properties of 1

- Parity:
- Rationality:
- Divisibility:
- Primality:

Inequalities

For any real numbers x and y , exactly one of the followings is true:

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Definition 5. Let x be a real number. Then

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|-----------------------------|---------------------------------|
| • x is positive if | • x is non-positive if |
| • x is negative if | • x is non-negative if |

Example 6.

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| • What is the negation of " $x < 3$ "? | • What is the negation of " $x \leq 5$ "? |
| • What is the negation of " $x > 4$ "? | • What is the negation of " $x \geq 6$ "? |